

Portfolio - Individual

AI-Systems Minor

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1 Student Profile

My passion for technology started loong before I began my studies at Fontys University of Applied Sciences. From assembling computers to troubleshooting them, I was always trying to learn more and more about how computers work. This curiosity rapidly turned into a broader interest in the digital world and the many possibilities it offers. Once I joined this university, I discovered a whole new dimension of computing: the field of artificial intelligence and data mining. What I found most interesting is how giving the computer a dataset, with some proper training, we can see all sorts of predictions and correlations, most of the time them being accurate.

In the past years, artificial intelligence has gained significant traction and attention from people who have integrated it into their daily routine, from using it to get a recipe for a quick meal to helping them in school assignments. But one task that AI has made considerably easier, is working with large amounts of data. In the past we would have to manually look over the data, analize and recognise patterns and then after come up with a conclusion. This all changed when the process became automated. For this reason, I decided to make use of this technology to come up with an application that can analyze a person's face and detect what is his mood based on facial expression. I think this application could have an impact across multiple domains, such as well-being monitoring, online recommendation systems and online learning.

1.1 Self reflection

During my studies at Fontys University, I've studied many topics which focus on programming, algorithms and data structures. During the 4th semester, I discovered a course that captivated my full attention from the first lesson, that being Data Mining. We discussed many topics like how we should handle large sets of data, many different regression models and neural networks. In this course we also had to do weekly assignments. My favorite one was an exercise where we had to predict house prices based on some features such as number of rooms, square feet and if it has a garade or not using a suited regression model. This couse sparked my curiosity in this domain and thats why I chose this minor.

1.2 SWOT Analysis

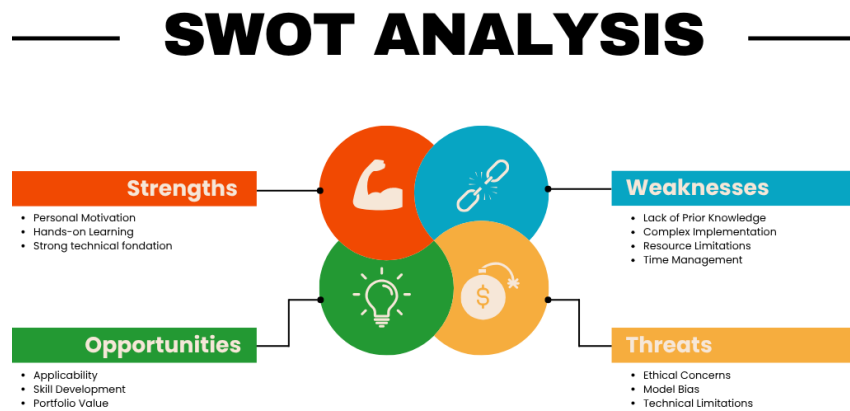


Figure 1: SWOT Analysis

Strengths

The project builds on my personal motivation and curiosity for AI, giving me a strong drive to complete it successfully. It also offers a hands-on learning experience where I can apply my technical knowledge in a practical context. My growing foundation in machine learning and computer vision supports the technical development of the model.

Weaknesses

Because I am still gaining experience in AI and computer vision, the project comes with a learning curve. The implementation can be complex, and limited time or computing resources may slow progress. Managing these aspects effectively will be essential to stay on schedule and maintain project quality.

Opportunities

This project allows me to explore a field with real-world applications, from well-being monitoring to adaptive learning systems. It provides the opportunity to strengthen my technical skills, learn new tools, and create something that can be showcased in my professional portfolio.

Threats

Potential risks include ethical concerns related to privacy and fairness, as well as the possibility of model bias if the dataset lacks diversity. Technical limitations such as lighting, camera quality, or real-time performance may also affect the system's accuracy.

2 Personal Learning Objectives

2.1 Learn about computer vision and machine learning

One of my main objectives for this project is to improve my ability to work with computer vision models, not just by studying their theory but by applying them to real-world image data. At the moment, I have some understanding of concepts such as convolutional neural networks and transfer learning, but I do not fully grasp how to implement them effectively for a task like mood detection. Through this project, I want to explore how to prepare and preprocess facial expression datasets, how to train and fine-tune different models, and how to evaluate their performance across various scenarios. By experimenting with multiple approaches and comparing their results, I aim to gain a more concrete understanding of how computer vision techniques can be used to build reliable systems that interpret human emotions.

2.2 Reflect on ethics of AI

During this project, I am hoping to expand my understanding of the implications of AI systems, primarily on the implications of AI systems on the recognition of facial emotion. I know some of the ethical limitations of AI, such as biases and privacy issues, but I have not had the opportunity to think on how they are relevant to my work. I hope to identify some of the risks and limitations concerning the fairness and accuracy of the system on different users and across multiple situations. This will be the first step in making my approach responsible, deriving a balanced and realistic understanding that recognizes the technical side of a project and the ethical concerns that must be considered. I hope to advance my work in a more responsible way within the social context

2.3 Managing study time

One problem that has frequently appeared during my studies, is that I tend to manage my time poorly and sometimes barely have any time to study. I tend to underestimate the workload and find myself in a difficult situation when there are deadlines involved. Although I am motivated, this habit tends to affect the way I work and my productivity. In this project I want to adopt a new strategy, by setting weekly deadlines and reserving 1-2 hours of self study per day. In my opinion this will help me set realistic milestones and prevent me from procrastinating and doing all the work in one day. This will not

only help me successfully complete this project, but will also prove an useful skill in further academic and professional work.

3 Context

The inspiration for this project comes from a childhood curiosity. Since I was a little kid, I always was fascinated by Apple's products and how they were two steps ahead of their competition. In September 2017, when they introduced the iPhone X, they also announced a new feature that amazed and still amazes me to this day. That feature was facial recognition, the possibility to unlock your phone just by looking at it. That moment sparked my curiosity about how machines can recognize and interpret human faces. At first, I was impressed by the security aspect of unlocking a device with facial recognition, but later I started wondering whether similar technology could also understand more subtle aspects, such as a person's mood or emotional state. This thought stayed with me, and as I learned more about artificial intelligence during my studies, I realized that computer vision and machine learning could make this possible. My project idea grew out of this interest: if a phone can identify who you are, perhaps a system can also detect how you feel, and this could open the door to meaningful applications in areas like well-being, online learning, and personalized recommendations.

4 Theoretical Learning Objectives

Up to this point, I have little knowledge about machine learning and computer vision. To be able to build a reliable model that can recognize emotions in real time, I still need to build up my knowledge in these areas and practice by working on large datasets available on the internet.

4.1 Main Research Question

How effectively can computer vision and machine learning techniques detect and classify a person's mood based on facial expressions captured from images, videos, or real-time camera input?

4.2 Sub Research Questions

1. How does the ethnicity of a person affect the performance and fairness of mood detection models?
2. How does real-time camera input compare to static images or pre-recorded videos in terms of accuracy and usability?
3. What ethical considerations need to be taken into account when developing and applying facial emotion recognition systems?

4. How does the model's accuracy in mood detection vary across different emotion categories?

A (sub)question is considered complete once I have gathered sufficient results, analysis, or reflection to confidently answer it based on evidence from my research and development process.

4.3 Topics

In this project, I plan to explore the following areas:

1. **Computer Vision:** I will learn how to detect and analyze faces from images and videos to recognize facial expressions.
2. **Machine Learning:** I will use machine learning models, such as convolutional neural networks, to classify emotions based on facial expressions.
3. **Datasets and Data Handling:** I will work with existing facial expression datasets and prepare them for training by cleaning, organizing, and analyzing their quality.
4. **Fairness and Ethics:** I will study how different factors, such as ethnicity, can affect the model's accuracy and how to ensure the system is fair and respects privacy rules.

4.4 Bloom's Taxonomy

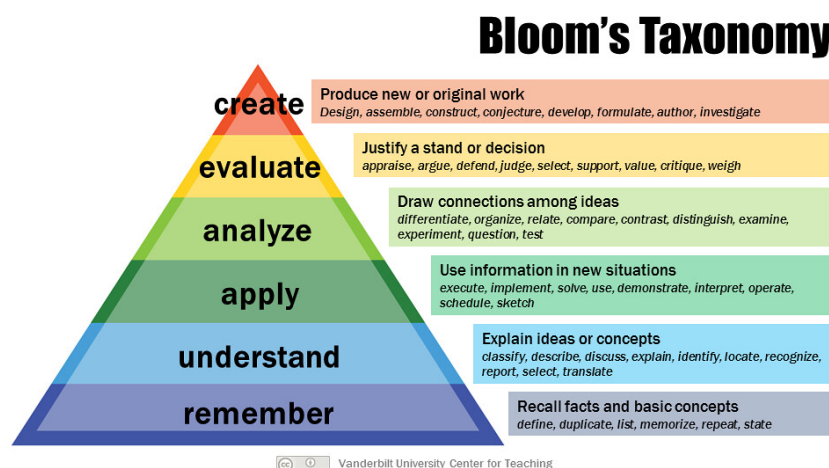


Figure 2: Bloom's Taxonomy

Computer Vision – Apply/Understand

By applying and understanding computer vision principles, I can learn how images are processed and how various algorithms detect and interpret facial features.

Machine Learning – Create/Analyze

This objective is a combination between Create and Evaluate. I aim to apply machine learning techniques to classify moods and then analyze how different models perform on specific datasets. By comparing their accuracy and limitations, I can determine which approach is more suitable for emotion detection.

Datasets and Data Handling – Analyze/Create

This objective goes hand in hand with Analyze and Create. By analyzing and preparing datasets to ensure reliability, I can determine if the available datasets on the internet are suitable for training mood detection models. This involves examining data quality and recognizing potential biases that may influence the results. After training the AI on datasets from the internet, I can create my own datasets by taking pictures of my friends expressing different moods and annotating them.

Fairness and Ethics – Evaluate

This objective focuses on evaluating the fairness and ethical implications of facial emotion recognition. I want to explore how differences in ethnicity may influence the accuracy and reliability of the model, and investigate potential biases that could arise from unbalanced datasets. In addition, I plan to study existing guidelines and policies for responsible AI development, so I can reflect on the standards that should be followed to ensure fairness, inclusivity, and ethical use of mood detection systems.

5 Learning Strategy

To be able to answer the research questions and reach my proposed learning objectives, I will apply several learning strategies:

1. **Desk Research:** Recap the lessons from the Data Mining course (semester 4) to strengthen my knowledge of machine learning and neural networks.
2. **Online Resources:** Study articles and watch YouTube tutorials on facial recognition and computer vision to support setting up my model.
3. **Fontys Lessons:** Attend the Deep Learning and Computer Vision lectures and prepare questions to ask lecturers afterwards.
4. **Development:** Experiment with different datasets and models to gain hands-on experience and develop the desktop app for presenting emotions in a user-friendly way.

Sub-Research Question	Strategy (including Learning Strategy)
How does the ethnicity of a person affect the performance and fairness of mood detection models?	Evaluate model performance across diverse datasets, analyze potential biases, and compare accuracy results between different ethnic groups. Learning Strategy: Desk Research (study bias and fairness in AI), Development (experiment with diverse datasets).
How does real-time camera input compare to static images or pre-recorded videos in terms of accuracy and usability?	Test the model on live webcam input and compare results with static images and pre-recorded videos to measure both accuracy and responsiveness. Learning Strategy: Development (implement experiments), Online Resources (tutorials on real-time computer vision).
What ethical considerations need to be taken into account when developing and applying facial emotion recognition systems?	Conduct desk research on AI ethics and legal frameworks (e.g., GDPR, EU AI Act) and reflect on responsible use, privacy, and fairness in application. Learning Strategy: Desk Research (policies and ethics), Fontys Lessons (ask questions on ethical AI).
How does the model’s accuracy in mood detection vary across different emotion categories?	I will evaluate per-class accuracy using confusion matrices to identify which emotions are most accurately recognized and where misclassifications occur. Learning Strategy: Development (evaluate model performance using confusion matrices), Desk Research (study performance metrics and evaluation methods).

Table 1: Sub-Research Questions with Strategies and Learning Strategies

6 Theoretical Learning Objectives

The main question to answer in this project is as follows:

How effectively can computer vision and machine learning techniques detect and classify a person's mood based on facial expressions captured from images, videos, or real-time camera input?

To answer this question, first we need to answer the subquestions mentioned below:

6.1 First Sub Question:

How does the ethnicity of a person affect the performance and fairness of mood detection models?

Introduction to Algorithmic Bias

Computer vision models are not inherently neutral; they reflect the data they are trained on. In the context of facial emotion recognition (FER), the ethnicity of a subject significantly impacts model performance. If a dataset is unbalanced (containing mostly images of one demographic group), the model will "overfit" to the facial features, skin tones, and lighting conditions typical of that group, while failing to generalize to others. This phenomenon is known as Algorithmic Bias.

The Dataset Problem: FER2013 vs. Reality

The model developed in this project was initially trained on the FER2013 dataset. Research indicates that FER2013, like many early academic datasets, is heavily skewed towards White/Western faces. This can lead to a number of problems:

1. **Feature Overfitting:** The MobileNet model learns to detect emotions based on specific landmarks (e.g., the curve of a mouth, the crinkle of an eye). If the training data lacks diversity, the model may treat features common to specific ethnicities (such as different eye shapes or brow structures) as indicators of specific emotions rather than neutral physical traits.
2. **Performance Gap:** Consequently, when the model is tested on non-Western faces (e.g., Asian or Black individuals), the accuracy drops. The "recall" score (The ability to find all examples of an emotion) is significantly lower for minority groups because the model simply hasn't seen enough examples to recognize their expressions confidently.

The "Angry Black Man" Paradox

One of the most documented and harmful failures in facial recognition fairness is the "Angry Black Man" paradox. This occurs when a model disproportionately misclassifies "Neutral" or "Serious" expressions of Black men as "Angry."

- **Cause:** The model learns to associate darker skin tones or specific natural resting face structures with the label "Angry" because the few examples of Black individuals in the training set might have been biased towards negative emotions, or simply because the model confuses physical features with emotional signals.

- **Impact:** In a real-world deployment (e.g., an automated interview screener or a security system), this bias could lead to unfair penalization or profiling of Black individuals, flagging them as aggressive when they are simply neutral. This violates the fairness principle outlined in the EU AI Act "Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence", 2021.

The "Polite Smile" Paradox

A significant paradox affects Asian demographics, often rooted in Cultural Display Rules rather than just physical features. This paradox occurs when models incorrectly classify expressions of embarrassment, anxiety, or politeness as "Happy."

- **Cause:** Western-centric datasets like FER2013 almost exclusively annotate the geometric feature of an upturned mouth as "Happiness." However, in many East Asian cultures (e.g., Japan, Thailand), smiling is frequently used as a social tool to mask negative emotions like embarrassment or confusion. Because the model lacks this cultural context, it sees the geometry of a smile and ignores other subtle cues (like the eyes) that might indicate distress.

- **Impact:** This bias is critical in healthcare or wellbeing applications. For example, an AI therapist or mental health monitoring app might incorrectly classify a distressed Asian user as "Happy" or "Doing well" simply because they are smiling out of polite discomfort. This leads to a false positive for well-being, potentially denying the user necessary help.

Mitigation Strategies and AffectNet

To address these ethnic disparities, simply training for more epochs is insufficient. The underlying data must be corrected:

1. **Dataset Balancing:** Moving from FER2013 to larger, like AffectNet provides a more balanced representation of global demographics, reducing the performance gap.
2. **Bias Auditing:** Running a "Bias Audit" (testing the model on a balanced dataset) is crucial to identifying and quantifying these risks before deployment.

Conclusion

Ethnicity plays a critical role in the reliability of mood detection models. Without intentional intervention, models trained on standard public datasets will exhibit significant bias, performing accurately on White faces while misinterpreting or under-performing on minority groups. Ensuring fairness requires a shift from model-centric optimization to data-centric curation, ensuring all ethnicities are represented equally in the training phase.

6.2 Second Sub Question:

How does real-time camera input compare to static images or pre-recorded videos in terms of accuracy and usability?

The performance of emotion detection systems differs noticeably on static images, pre-recorded videos and real-time camera input. During the development and testing of the desktop application the differences were clearly observed in both accuracy and usability when using real-time webcam input compared to controlled data, such as pre-recorded videos and images.

When using static images, the emotion detection model achieved the highest accuracy. Static images are usually well-aligned, centered and captured under controlled light conditions. In most of the public datasets, such as FER2013 and AffectNet, facial expressions are exaggerated and clearly labeled, which makes the job for the model much easier. Because of the high-quality frame, there is no motion blur or occlusion and the predictions tend to be more stable.

Pre-recorded videos tend to be the middle ground between static images and real-time input. Videos provide multiple frames of the same expression, allowing the model to use more complex algorithms, like majority voting. However, pre-recorded videos are still most of the time captured in controlled environments, often showing clear facial expressions. As a result, while performance is still lower than a static images due to motion blur and lighting inconsistencies, accuracy tends to remain relatively high and consistent. Real-time camera input introduces several challenges that affect accuracy. In real life, facial expressions are often subtle, incomplete or mixed. While testing the desktop application, it was observed that the model struggled with detecting mild frustration or subtle anger. These emotions were sometimes misclassified as NEUTRAL rather than NEGATIVE. This is a clear example of why emotion training datasets often do not reflect real life scenarios, because in the datasets, frustration is often showed by wide eyes or exaggerated frowns, but in real life, frustration does not always involve strong facial cues. Another interesting observation is that this issue occurred far less for the POSITIVE class. Positive emotions such as happiness tend to produce clearer and more distinctive facial features, such as smiles or raised cheeks, which are easier to detect in real-time conditions. This resulted in stable and accurate detection for the POSITIVE class.

From an usability perspective, real-time camera input is much more challenging but also

more realistic. While static images and pre-recorded videos are useful for training and benchmarking, they do not reflect the real world situations. Real-time input captures the face in different light conditions, has to take into account head movements, partial occlusions and spontaneous expressions. All of these factors reduce the classification accuracy but increase external validity. In other words, real-time performance is a more meaningful indicator of how the system is going to perform in practical applications.

Conclusion

Overall, static images provide the highest accuracy but the lowest realism, pre-recorded videos offer a balance between accuracy and temporal context, and real-time camera input presents the lowest accuracy due to subtle and natural expressions but the highest usability in real-world scenarios. The observed misclassification of subtle negative emotions as neutral highlights an important limitation of emotion recognition models trained on staged datasets and reinforces the decision to use broader emotion categories in the final application.

6.3 Third Sub Question:

What ethical considerations need to be taken into account when developing and applying facial emotion recognition systems?

6.3.1 Privacy and Data Protection

Facial data constitutes sensitive biometric information. To minimize risk, the project follows data minimization and purpose limitation, as required by the GDPR “General Data Protection Regulation (EU) 2016/679”, 2016.

- **Local processing:** All inference is performed on-device; no raw images or identifiers are transmitted.
- **No permanent storage:** By default, images/frames are processed in memory only. Any diagnostic logging excludes personal data.
- **Data deletion:** Temporary buffers are cleared immediately after inference.

6.3.2 User Consent and Transparency

The application implements explicit, informed consent in compliance with Article 6(1)(a) of the GDPR “General Data Protection Regulation (EU) 2016/679”, 2016.

- **Opt-in:** Camera access and emotion inference are disabled until the user enables them.

- **Just-in-time notices:** A short notice explains what is processed, for what purpose, and for how long.
- **Controls:** Clear on/off toggle, session indicator (e.g., icon/LED), and a link to the privacy summary.

6.3.3 Data Storage and Security

If short-term storage is required for evaluation, the design follows the principles of integrity and confidentiality from Article 5 of the GDPR “General Data Protection Regulation (EU) 2016/679”, 2016 and ISO-style information security best practices.

- **Scope:** Store only what is strictly necessary (e.g., aggregated metrics).
- **Protection:** Encrypt at rest; restrict access to authorized personnel.
- **Retention:** Define retention windows and automatic deletion schedules.

6.3.4 AI Bias and Fairness

The model may inherit bias from training data, which is addressed in line with the fairness principles of the EU AI Act “Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence”, 2021. Mitigations include:

- Inspecting class distribution and augmenting underrepresented classes.
- Reporting class-wise metrics and confusion matrices.
- Avoiding high-stakes use; keeping predictions informational, not determinative.

6.3.5 Ethical Use and Context

The main goal of this project is to use emotion detection in a positive and helpful way, such as reminding users to take breaks or encouraging them when they show positive emotions. The system is not made for surveillance, profiling, or influencing people’s behaviour. All demonstrations and tests clearly explain the purpose of the system to anyone involved. Users have full control over when the application is active and can pause or stop it at any time. Any future version of the app will show a clear indicator when the camera is on and will include a short, easy-to-understand privacy explanation. By setting clear limits and responsible uses, the project aims to keep emotion detection ethical and focused on helping people, consistent with the IEEE’s principles of human wellbeing and accountability “Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems”, 2019.

6.3.6 Regulatory Alignment

The project aligns with key European data-protection and AI-ethics frameworks, including the General Data Protection Regulation (GDPR) “General Data Protection Regulation (EU) 2016/679”, 2016 and the upcoming EU AI Act “Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence”, 2021. Under GDPR, it respects all major principles: lawfulness, fairness, transparency, purpose limitation, data minimisation, and integrity/confidentiality. The design treats all facial analysis as sensitive processing and ensures that it stays within a low-risk category by design—local processing, explicit user consent, and transparent operation. In terms of governance, documentation is maintained for the model’s purpose, assumptions, and limitations to promote accountability. The project also draws on the IEEE Ethically Aligned Design framework “Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems”, 2019, focusing on human agency, accountability, and the promotion of wellbeing. Together, these measures ensure that the prototype not only complies with current legal requirements but also anticipates future standards for trustworthy and responsible AI systems.

Risk & Mitigation Summary

Risk	Description	Likelihood	Mitigation
Privacy leakage	Unintended storage/transmission of frames	Low	On-device processing, no raw logs, code review
Bias	Uneven accuracy across groups/classes	Medium	Class-balance checks, per-class metrics, iterative tuning
Misuse	Use for surveillance without consent	Low/Med	Usage policy, UI indicators, opt-in only
Security	Unauthorized access to temp data	Low	Encryption, least-privilege access, short retention

6.3.7 Conclusion

With local processing, explicit consent, limited storage, fairness checks, and clear scope limits, the prototype adheres to responsible AI practices and prepares for compliant, user-respecting experimentation. This approach demonstrates alignment with the ethical frameworks of GDPR “General Data Protection Regulation (EU) 2016/679”, 2016, the EU AI Act “Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence”, 2021, and IEEE Ethically Aligned Design “Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems”, 2019.

6.4 Fourth Sub Question:

How does the model’s accuracy in mood detection vary across different emotion categories?

To answer this question, three iterations of the model were evaluated:

1. **Model A:** 7 Classes (All emotions), trained for 25 Epochs.
2. **Model B:** 7 Classes (All emotions), trained for 35 Epochs.
3. **Model C:** 3 Classes (Positive, Negative, Neutral), trained for 35 Epochs.

The performance of each model was analysed using the per-class F1 scores and confusion matrices. These reveal how reliably the system can distinguish between specific emotions and which categories remain challenging for the MobileNet architecture trained on FER2013.

Model A: 7 Classes, 25 Epochs

Model A shows a clear imbalance in performance across emotion categories. Emotions with highly expressive and distinct facial cues perform best, while subtle or ambiguous categories remain problematic.

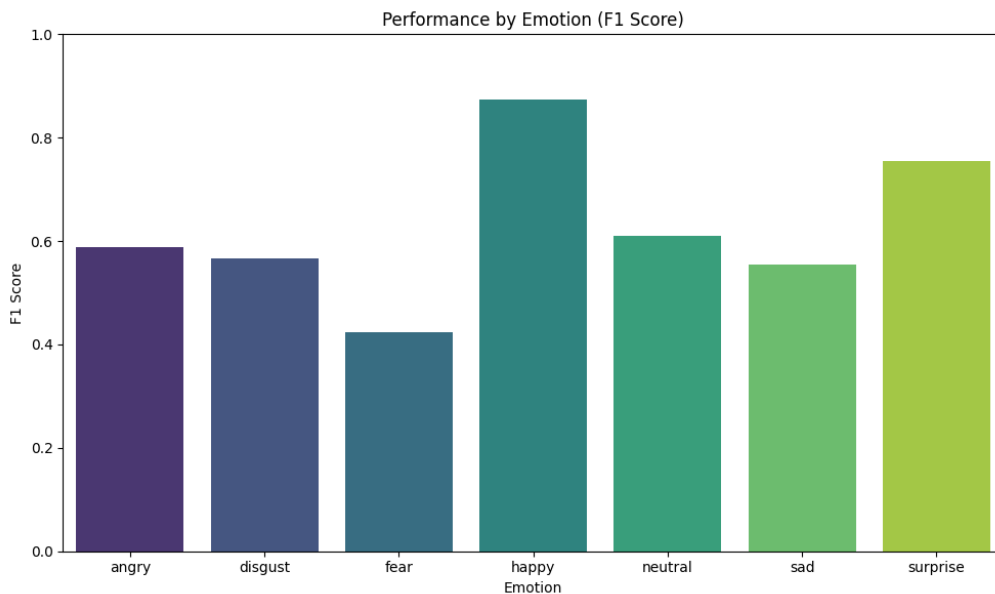


Figure 3: Model A — F1-score per class (7 classes, 25 epochs)

- **Highest-performing emotions:** *happy* ($F1 \approx 0.87$) and *surprise* ($F1 \approx 0.76$). These emotions have strong visual signals and are well-represented in FER2013.
- **Moderate performance:** *angry* ($F1 \approx 0.59$), *neutral* ($F1 \approx 0.61$), *disgust* ($F1 \approx 0.57$), and *sad* ($F1 \approx 0.55$). These emotions often overlap visually, leading to frequent misclassifications.

- **Lowest-performing emotion:** *fear* ($F1 \approx 0.42$), which is heavily confused with *sad*, *surprise*, and *angry*. This aligns with the literature, as fear expressions are subtle and inconsistently depicted in FER2013.

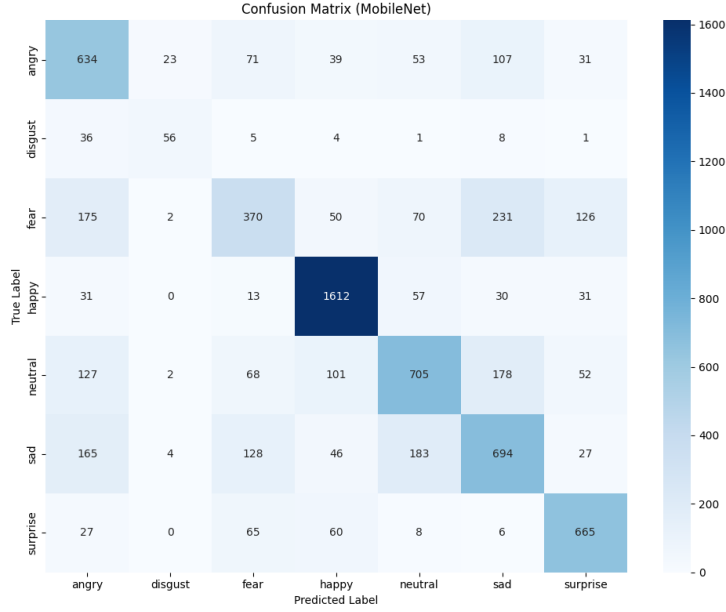


Figure 4: Model A — Confusion Matrix (7 classes, 25 epochs)

The confusion matrix confirms broad dispersion among negative emotions, suggesting the model struggles to differentiate fine-grained emotional cues under limited training duration.

Model B: 7 Classes, 35 Epochs

Increasing the training time to 35 epochs improves overall stability and class separation.

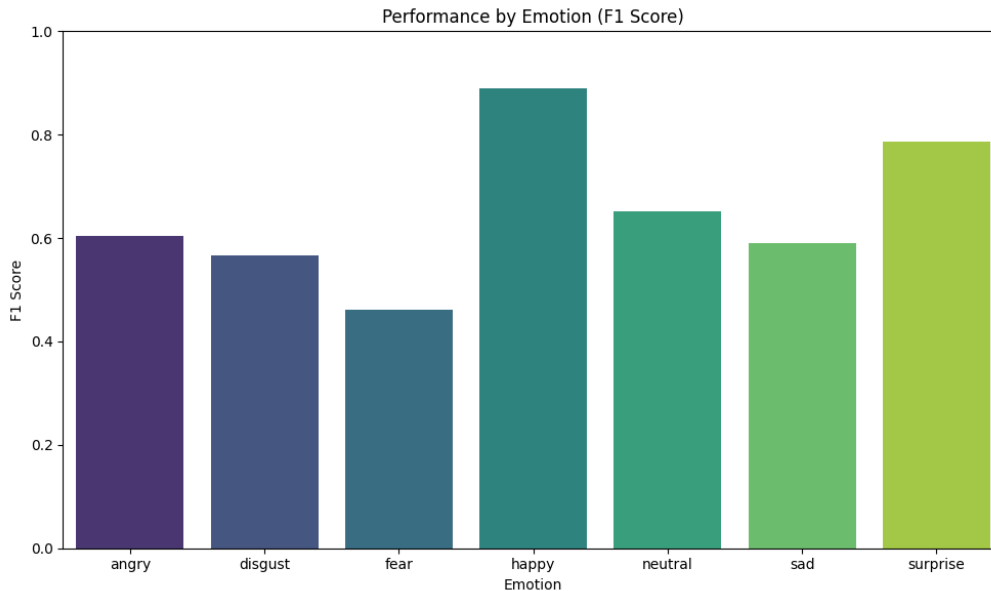


Figure 5: Model B — F1-score per class (7 classes, 35 epochs)

- **Improvements:** Most emotion categories show higher F1 scores compared to Model A. Notably, *happy* remains the strongest class ($F1 \approx 0.89$), and *surprise* increases to roughly 0.79.
- **Negative emotion patterns:** *angry*, *disgust*, *sad*, and *neutral* all experience modest improvements, but the confusion matrix still reveals strong overlap among these categories.
- **Fear remains challenging:** Even with extended training, *fear* continues to have the lowest performance ($F1 \approx 0.46$), confirming its status as the least distinguishable category.

Overall, Model B demonstrates that longer training helps refine class boundaries, but the inherent ambiguity and dataset imbalance in FER2013 limit the achievable accuracy for certain emotions.

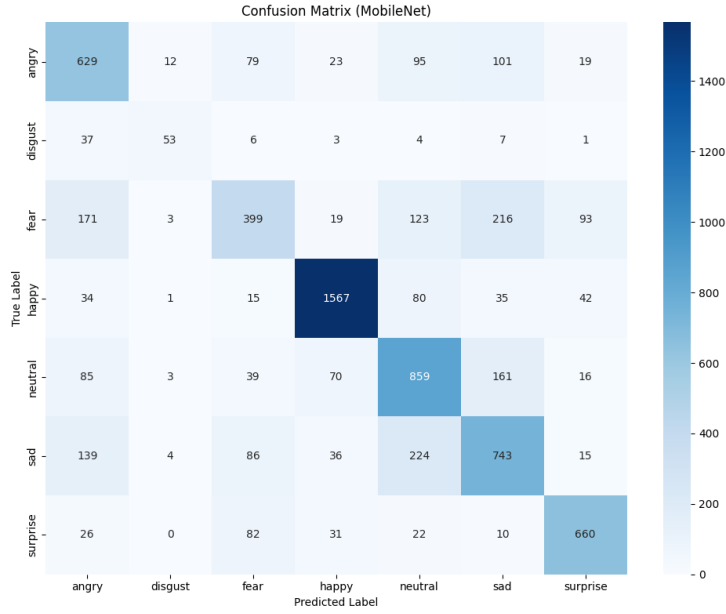


Figure 6: Model B — Confusion Matrix (7 classes, 35 epochs)

Model C: 3 Classes (Positive, Negative, Neutral), 35 Epochs

When emotions are grouped into broader categories, performance improves substantially. This setup reduces the complexity of distinguishing subtle expressions and mitigates dataset imbalance.

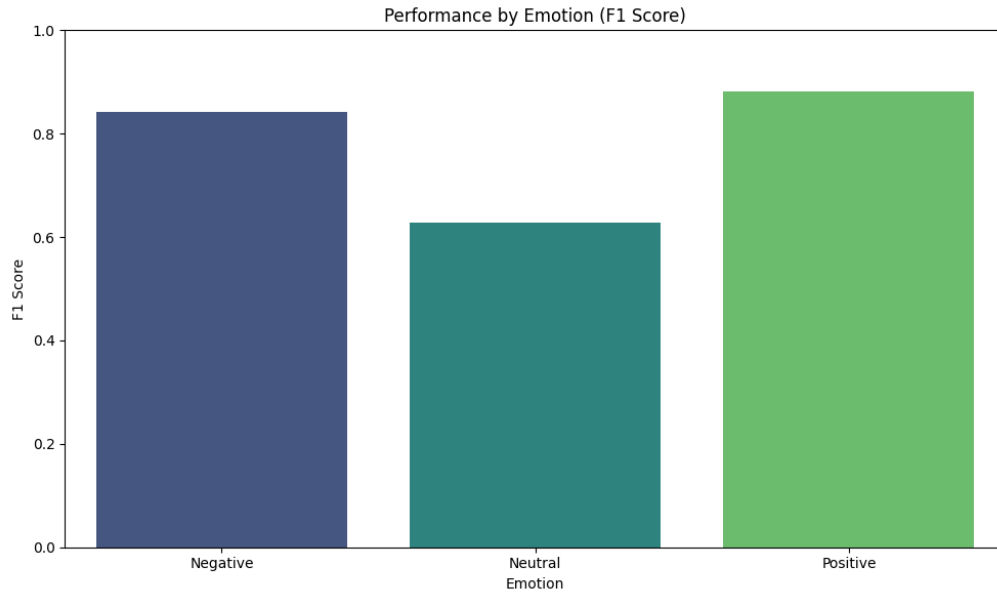


Figure 7: Model C — F1-score per class (3 classes, 35 epochs)

- **Positive:** The strongest category ($F1 \approx 0.88$), driven largely by the clarity of the *happy* class.
- **Negative:** Strong performance ($F1 \approx 0.84$), showing that grouping multiple neg-

ative emotions stabilises classification.

- **Neutral:** The weakest category ($F1 \approx 0.63$), frequently misclassified as Positive or Negative. This reflects the inherent ambiguity of neutral expressions and their overlap with low-intensity emotions.

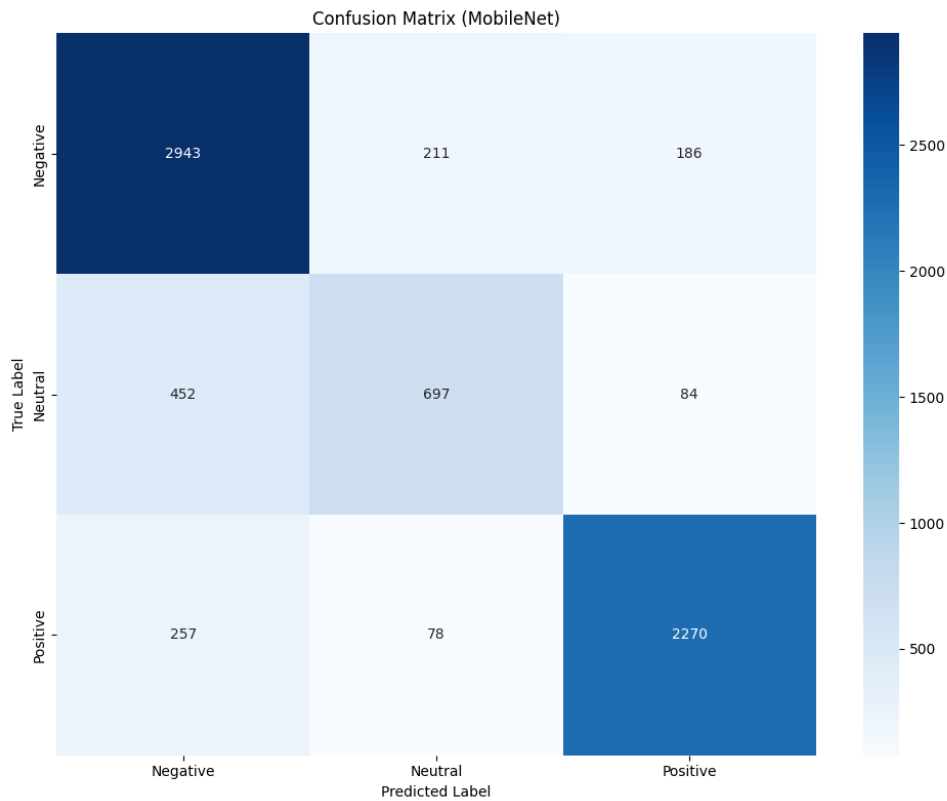


Figure 8: Model C — Confusion Matrix (3 classes, 35 epochs)

The confusion matrix shows that Neutral expressions often sit between Positive and Negative cues, explaining the reduced F1 score compared to the other two classes.

Conclusion

Across all three models, certain trends are consistent:

- Emotions with clear, distinct facial features (e.g., *happy*, *surprise*) consistently achieve the highest accuracy.
- Subtle, low-intensity, or culturally variable emotions (*fear*, *disgust*, *sad*) remain challenging regardless of training duration.
- Increasing epochs improves performance but does not fully resolve dataset-driven limitations.
- Reducing the classification task to broader emotion groups significantly increases accuracy, especially for Positive and Negative categories.

In summary, the model's accuracy varies substantially across emotion types, with high performance on expressive emotions and weaker performance on subtle or ambiguous categories. This variation reflects both the limitations of FER2013 and the biological reality that some emotions are inherently harder to recognise.

7 Apporach

7.1 Picking a dataset

I choose the FER2013 dataset because it offers a practical and accessible starting point in developing an emotion detection model. Because this dataset is widely used in academic projects, the data is already pre-labeled, easy to load and works well with popular deep-learning libraries, which made the experimentation much faster. Even though the dataset has its limitations (low resolution images and some class imbalance), it still provides a solid base for understanding how facial emotion recognition works. Being the most convenient and beginner-friendly option, I considered that training a model on this dataset, I can obtain a good proof of concept and maybe later on move forward on a much bigger and demographic diverse dataset (eg., AffectNet).



Figure 9: Showcase of all the classes in FER2013 dataset

7.1.1 Altering the dataset

After analysing the confusion matrices and F1-scores of models A and B (7 classes, 25 epochs and 25 epochs), I saw that several classes showed low separability and high misclassification rates ("False Positives"). Emotions such as *fear*, *sad* and *disgust* showed strong overlap, due to the similar features (frowning eyebrows, open mouth, wide open eyes), while *neutral* class overlapped with both positive and negative classes. These patterns show a clear limitation of the FER2013 dataset, where subtle emotions are difficult to classify.

Even after increasing the training time from 25 epochs to 25 epochs, the model still struggled in this area. After analysing the dataset and the project itself, it came to my mind that I did not need to know exactly how the end user felt, but just to now if he is in a bad or good mood. That's where I decided to narrow the classes and make them broader. Thus the emotions were grouped into 3 broader categories:

- **Positive** class that contained the *happy*, *surprise* classes.
- **Negative** class that contained the *angry*, *sad*, *fear*, *disgust* classes.
- **Neutral** class that contained the *neutral* class.

7.2 Training the model in Google Colab

Because of the size of the dataset and the limited hardware at my disposal, I choose to train my model on Google Colab.

Google Colab is a hosted Jupyter Notebook service that provides free access to computing resources, including top-tier GPU's (eg., NVIDIA T4 GPU, GOOGLE v5e-1 TPU). Using Google Colab I managed to cut the training time by almost 3 times (1.2 hours on Google Colab vs. 3.5 on my personal GPU). After the training is done, the h5 file (The file that contains the weights and model configuration) is stored in the drive.

7.3 Result Analysis

For the result analysis, the following performance indicators are considered:

- **Precision**
- **Recall**
- **F1-score**
- **Training/Validation Loss**
- **Training/Validation Accuracy**

Across all three models, these metrics provide insight into both the quality of the predictions (Classification reports) and the training behaviour (learning curves over multiple epochs). The following subsections present the results for each model individually before comparing them.

Model A: 7 Classes, 25 Epochs

The first version of the model was trained for 25 epochs on all seven FER2013 emotion categories. This configuration provides the baseline for evaluating later improvements.

Class	Precision	Recall	F1-score
angry	0.58	0.63	0.60
disgust	0.57	0.55	0.57
fear	0.43	0.45	0.43
happy	0.88	0.90	0.88
neutral	0.61	0.63	0.61
sad	0.55	0.59	0.56
surprise	0.75	0.78	0.75
Accuracy		0.66	
Macro Avg.	0.62	0.65	0.63
Weighted Avg.	0.66	0.66	0.66

Table 2: Classification performance for Model A (7 classes, 25 epochs).

Training Behaviour:

The training curves indicate that the model learns steadily, with validation accuracy stabilising around 0.62–0.65. However, training accuracy fluctuates heavily, suggesting instability caused by the complexity of distinguishing seven emotion categories in low-resolution images. Validation loss does not reduce significantly after the midpoint of training, confirming that the model reached its learning limit early.



Figure 10: Model A — Training and validation accuracy and loss (7 classes, 25 epochs).

These results highlight two key issues: (1) high confusion between negative emotions (angry, fear, sad), and (2) the dominance of the “happy” class, which consistently achieves the highest scores.

Model B: 7 Classes, 35 Epochs

Model B extends training to 35 epochs to observe whether longer optimisation improves performance.

Class	Precision	Recall	F1-score
angry	0.60	0.64	0.61
disgust	0.57	0.56	0.57
fear	0.47	0.47	0.46
happy	0.88	0.89	0.88
neutral	0.65	0.66	0.65
sad	0.58	0.60	0.59
surprise	0.78	0.80	0.78
Accuracy		0.69	
Macro Avg.	0.64	0.66	0.65
Weighted Avg.	0.69	0.69	0.69

Table 3: Classification performance for Model B (7 classes, 35 epochs).

Training Behaviour:

The training curves reveal a much smoother learning process compared to Model A. Validation accuracy increases gradually and reaches ≈ 0.68 , while validation loss decreases consistently.

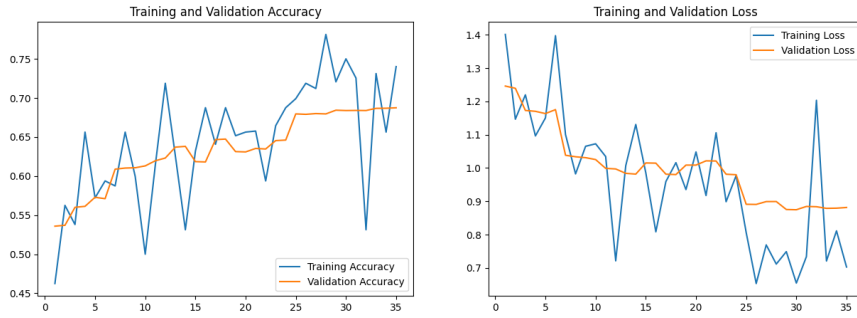


Figure 11: Model B — Training and validation accuracy and loss (7 classes, 35 epochs).

The improvement from 25 to 35 epochs is clear:

- Negative emotions receive slightly higher F1-scores.
- “Happy” remains the strongest and most separable class.
- “Fear” continues to be the hardest class to classify, showing chronic overlap with similar emotions.

Although the model improves with more training, the confusion between semantically similar emotions remains a limiting factor, illustrating the constraints of FER2013’s uneven and ambiguous labels.

Model C: 3 Classes (Positive, Neutral, Negative), 35 Epochs

To address the difficulty of distinguishing subtle expressions, the final model groups emotions into three broader categories. This reduces complexity and mitigates dataset imbalance.

Class	Precision	Recall	F1-score
Negative	0.81	0.88	0.84
Neutral	0.71	0.57	0.63
Positive	0.89	0.87	0.88
Accuracy		0.82	
Macro Avg.	0.80	0.77	0.78
Weighted Avg.	0.82	0.82	0.82

Table 4: Classification performance for Model C (3 classes, 35 epochs).

Training Behaviour:

Model C shows the cleanest and most stable training pattern. Validation accuracy increases steadily and reaches ≈ 0.82 , while validation loss decreases smoothly.

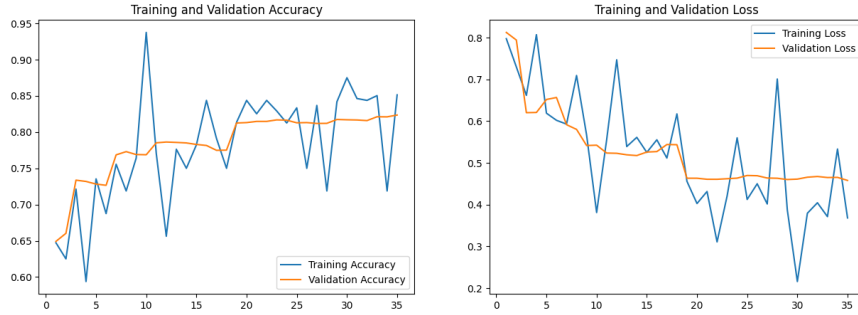


Figure 12: Model C — Training and validation accuracy and loss (3 classes, 35 epochs).

The improvement in performance is explained by:

- Clearer decision boundaries between broad categories.
- Reduction of label ambiguity (e.g., angry vs. sad vs. fear collapse into “Negative”).
- More stable gradients during training.

Neutral remains the weakest category due to its natural overlap with low-intensity positive and negative expressions.

Comparison of All Models

Overall performance improves significantly across the versions:

- **Model A (25 epochs, 7 classes):** Shows unstable training and modest accuracy (66%).
- **Model B (35 epochs, 7 classes):** More stable optimisation and improved accuracy (69%).
- **Model C (35 epochs, 3 classes):** Strong generalisation, clean training curves, and highest accuracy (82%).

The training curves clearly demonstrate that:

- Increasing epochs improves performance for the full 7-class model, but only to a limited extent.
- The FER2013 dataset makes fine-grained emotion classification difficult due to ambiguous labels, low image quality, and class imbalance.
- Grouping emotions into broader categories dramatically stabilises training and boosts accuracy.

8 Desktop Application: Study Buddy

8.1 Application Overview

To demonstrate the practical applicability of the emotion detection model (Model C), I developed a lightweight desktop application. The goal of the application is to support healthy studying habits by combining time management with context-aware mood feedback. The application runs in the background as a system tray tool and assists the user during study sessions by suggesting breaks or providing motivational feedback based on how the user is doing.

8.2 Application Design

The application is designed as a system tray application using PyQt6. Instead of a traditional window-based interface, I opted for a tray application to minimize distractions and reduce fatigue when switching tabs on your personal computer. If the user disables the camera, the application does not run inference anymore and acts only as a timer, giving alerts only when the study session is about to end.

Through the tray menu, the user can:

- Start, pause, resume or stop the study session.

- Select predefined session durations (25, 45, 60, 90, 120 minutes).
- Enter a custom session duration.
- Enable or disable the camera.
- Exit the application.

The session status and the remaining time are displayed dynamically in the tray tooltip and menu, giving users quick access to checking their progress without interrupting their workflow.

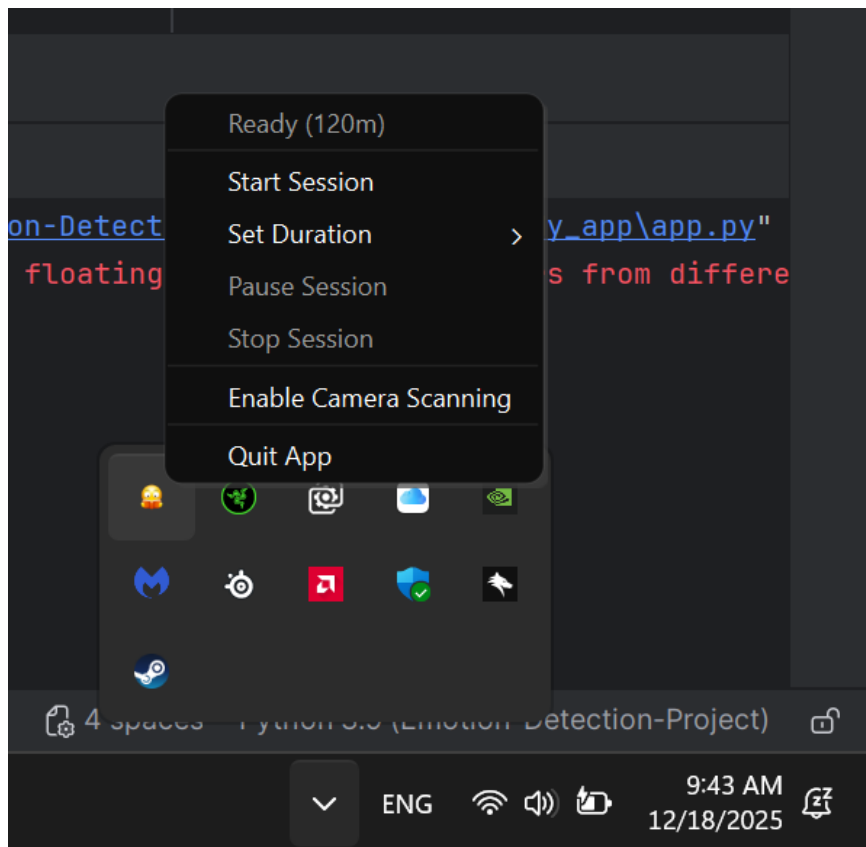


Figure 13: Preview of the application menu from the system tray

8.3 Study Session and Time Management

At the core, the applicaiton is a study session controller that tracks the time that you study. The application sens time-based notification, letting the user know the study session is comming to an end (60, 30, 10 minutes remaining). This timer functionality suports the application's primary purpose: helping the user structure their study time and avoid burnout and fatigue.

8.4 Mood Detection Pipeline

When the camera is enabled, the application is able to scan your face periodically and perform short emotion scans. These scans occur once per minute and run in the background thread to ensure the user interface remains responsive.

The emotion detection pipeline consists of the following steps:

- Webcam input: The frames are captured using OpenCV
- Face detection: A Haar cascade classifier is used to locate a face in each frame
- Preprocessing: The images captured from step one are resized to 224x224 pixels and normalized
- Model Inference: The preprocessed image is passed to the trained MobileNet model (Model C)
- Prediction Stabilisation: Multiple predictions are collected during a short scan and combined using a majority-vote strategy to reduce single-frame misclassifications.

8.5 Mood-Based Decision Logic

The mood-based decision logic is responsible for deciding when the application should interact with the user. Instead of responding immediately to every emotion detected by the camera, this logic evaluates the moods collected from the camera over time and determines whether the application should prompt a notification or not. This approach is the most realistic one because facial expressions can change quickly and single detections will not represent how the user is feeling during a longer study session.

Each time the application performs a scan, the camera stays open approximately 4 second, capturing multiple frames to produce a final mood classification. This classification can be neutral, positive or negative and represents the dominant facial expression during that scan. After a mood is produced, it is stored together with a timestamp at which it was detected. The results are kept in the RAM, representing the user's most recent emotional state. Only results from a predefined period are stored, while the older entries are automatically removed. This ensures that decisions are always based on current information and help keep the application as lightweight as possible.

Once a new mood had been added to the history, the decision logic analyses the stored data by counting how often each mood occurred within the period. The system looks for patterns that indicate a consistent emotional state rather than reacting to a single negative result. For example, if an user is in a positive state until the scan, but then his facial expression changes suddenly during the scan and gets classified as a negative mood, that is considered a misclassification and the app should not tell the user to take a break. This way the logic looks for repeated negative detections within a period to notify the user to take a break.

A break notification is triggered only when two conditions are met:

- The most recent mood detection must be negative, ensuring that the user is experiencing difficulty.
- The number of negative mood detections within the period must reached the pre-defined threshold (eg., 5 negative detections in a row).

With these 2 conditions combined, we avoid spamming the user with notifications by waiting until there is sufficient evidence that a break may be needed.

To prevent the application from interrupting the user too frequently, the decision logic has a cooldown mechanism. After a notification is shown, the system temporarily disables further notifications of the same type. During this cooldown period, new mood results are still recorded, but no additional messages are displayed. This design choice helps maintain a supportive and non-intrusive user experience.

When the recent mood history consists mainly of neutral or positive detections and no break suggestion has been triggered recently, the system may display a motivational message instead. These messages are intended to encourage the user to continue their study session and provide positive reinforcement without implying that the user is struggling.

8.6 Privacy-Aware Design

All image processing and emotion interference occur locally on the user's device. No images, video frames or personal information that may lead to identifying an individual are stored or transmitted. The webcam feature is fully opt-in and users can disable mood detection at any time via the tray menu.

The design aligns with GDPR and EU AI Act principles such as data minimisation and purpose limitation.

8.7 Limitations

While the application successfully demonstrates the integration of emotion recognition into a practical system, several limitations remain. The face detection relies on a Haar cascade, which may struggle under poor lighting conditions or with non-frontal faces. Additionally, the emotion model is trained on FER2013, which limits its robustness across diverse demographics. Finally, the application provides supportive feedback only and is not suitable for high-stakes or clinical use.

9 Reflection

When I started this project, I viewed creating a model as straightforward as feeding the dataset to the model and then giving it time to train. I assumed that if I give the FER2013 dataset to the Neural Network that I coded, my model will learn to "see" the emotions as a human would. At first, when I saw the accuracy from the "Happy" class

being as high as 87%, I thought that the model is perfect but just needs a little more time to train. After giving it extra 10 epochs, I saw that the model started to plateau, meaning that the loss function started flattening out, in normal words the model stopped learning. No matter what I did the "Fear" class just would not get a higher precision than 43%.

That was the breaking point where I considered that maybe the model was not the problem and took a step back to look through the images that the model trains on. I decided to quickly run through the dataset to see how many images I have per class to check for a potential class imbalance. After doing so, not only did it confirm my suspicions, I also for some classes could not tell myself what class they were. Seeing this, a genius idea sparked in my mind. I remembered from Data Mining classes that in one lesson we talked about graphs and how they give us insight in what happens during the training, so I decided to generate a **Confusion Matrix**, and that was the best decision for this project. I saw that the Negative classes got often confused with each other, for example the "Fear" class was getting confused with the "Sad" class 216 times.

That was the moment when I realized that my model was not dumb, it was just fighting to understand low resolution images with very similar features. That is when I stopped thinking as a "Software Engineer", but more like a "Data Scientist". I realized that trying to make the model accurately detect the 7 classes was just a losing battle. I found myself again taking a step back and returning to the initial project idea. The point of my model was to detect if the person using my application was not feeling well and needed a break from studying, not to detect how he was feeling. Then I tried to group all of the 7 classes into 3 simple classes: Negative, Positive and Neutral. That idea was the breaking point for my project. I managed to get a precision of 89% on "Positive" classes, 71% on "Neutral" classes and 81% on "Negative". After going through the "Neutral" photos to see why I got a relatively low Recall (57%) compared to the other classes, I saw that there are some cases where the images were really similar. So it was not a model problem, it was a dataset problem. After reaching an acceptable performance on the model, I started my research on the legal part of AI and all the ethical problems that emotion detection comes with.

At the start of this semester, I had a meeting with my project coach, Mr. Frank van Gennip. In that meeting we discussed about potential research subquestions. That meeting opened my mind to a layer of AI that I had completely neglected: the legal and the ethical part. Until that meeting I saw training a model as a technical challenge. Mr. van Gennip helped me realize that AI is also a legal and social responsibility. We discussed that an AI must also adhere to frameworks like GDPR, IEEE standards for ethical design and also the EU AI Act to avoid any potential problems with the law. I also learned that emotion recognition systems are often classified as "high risk" because they work with biometric data.

This new mindset that my coach helped me create led me to another critical subquestion: **How does a person's ethnicity affect the model?** Looking back to my earlier realization about the data quality, I looked again at the dataset, but from an ethical

perspective rather than a statistical one. The FER2013 dataset, like many open source datasets, had problems with demographic imbalances. That led to another problem, because if the model is trained on predominantly white faces, the features that it learns will not generalize well on other ethnicities. This is a real problem because I realized that my model was algorithmically biased. A model that works well for one demographic but fails for another violates the fairness principles written in the IEEE and EU AI Act. To sum up my journey throughout this minor, I realized that I started as a software engineer that was looking only for high precision and recall percentages. But in the end I ended up seeing over the lines of code and realize that code cannot ignore the law. Moving forward, I learnt that building an AI requires not only technical skills, but also empathy to understand the data and adhere to the legal frameworks to avoid legal problems.

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